

PROBLEM STATEMENT

NIRF Ranking Intelligence & Analytics Platform

Build a **web-based analytics and prediction platform** that extracts NIRF ranking data from official PDF reports, transforms it into structured form, performs **exploratory data analysis and statistical hypothesis testing**, and applies machine learning to **detect ranking bias, model feature importance, and forecast ranking trends** across institutions and years — surfacing insights completely invisible in raw PDF rankings.

FOUR CORE GAPS THIS SOLVES

 DATA LOCKED IN PDFS NIRF publishes no API or dataset. Extraction pipeline is the first real engineering problem.	 HIDDEN RANKING BIAS Users see ranks, not the systemic patterns favouring certain institution types.	 NO TREND VISIBILITY No tool lets institutions track rank movement or forecast future position over time.	 NO STATISTICAL PROOF Observations about ranking patterns are never validated with rigorous statistical tests.
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ML ALGORITHM DECISION MATRIX — ALL OPTIONS EVALUATED

XGBoost ★ BEST CHOICE Primary · Rank Modeling + Bias Detection Accuracy Explainability Missing Data Bias Analysis + Non-linear feature interactions (TLR×RPC etc.) + Native SHAP support for bias explanation + Handles noisy, messy real-world PDF data + Best accuracy on medium tabular datasets + Industry standard — strong in interviews → Final model. No other algorithm matches this balance of accuracy + explainability for this exact problem.	LightGBM STRONG ALT Alternative · Speed-Optimised Gradient Boost Accuracy Explainability Speed Bias Analysis + Faster training at large scale + Comparable accuracy ceiling to XGBoost - Less mature SHAP / bias tooling - Overkill for NIRF dataset size (~thousands of rows) - Harder to explain in evaluations → Use only if data scales beyond 100K rows. Irrelevant for NIRF scope today.	Random Forest BASELINE Comparison Baseline Model Accuracy Explainability Stability Bias Analysis + Low variance, robust to overfitting + Good feature importance baseline + Easy to train and interpret - Lower accuracy than boosting methods - Weaker at capturing feature interaction → Train this first. If XGBoost doesn't beat it by >5%, something is wrong in preprocessing.
Prophet / ARIMA TREND MODULE Time-Series Rank Forecasting + Purpose-built for yearly trend forecasting + Prophet handles missing years gracefully + ARIMA good for univariate rank series - Cannot do multi-feature bias modeling - Needs sufficient historical points per institution → Run as a separate module alongside XGBoost. Don't try to force XGBoost to do time-series.	Linear Regression BENCHMARK Naive Benchmark Only + Interpretable coefficients + Good sanity-check baseline - Assumes linearity — invalid for rankings - Cannot capture TLR × RPC interactions - Fails on non-Gaussian residual patterns → Include purely as benchmark. Expect it to underperform. That proves why XGBoost is needed.	SVM / KNN / Neural Net SKIP Not Suitable for This Problem - KNN fails on high-dimensional tabular data - SVM: no native feature importance - Neural nets: overkill, uninterpretable for this dataset size - None handle missing values natively → Theoretically possible. Practically wrong. Evaluators will ask why — the answer above is your defense.



Why XGBoost is the Definitive Choice

- NIRF is structured/tabular — XGBoost's native domain

→ Handles NaN from PDF extraction without imputation hacks

→ Trains in under 10 seconds on NIRF dataset size
- SHAP values directly power the bias analysis layer

→ Non-linear TLR × RPC interactions captured automatically

→ Feature importance explains which parameters drive rank the most

STATISTICAL HYPOTHESES TO TEST

H ₁	Top-ranked institutions have significantly higher RPC scores than lower-ranked peers	t-test
H ₂	IITs/NITs show statistically different rank distributions vs. private colleges	ANOVA
H ₃	Research score has stronger rank correlation than outreach score	Spearman ρ
H ₄	Urban institutions are disproportionately represented in top 20	Chi-square

TECHNOLOGY STACK

XGBoost	SHAP	Prophet	FastAPI	React
Recharts	pdfplumber	pandas	scipy.stats	
scikit-learn	PostgreSQL			

REALISTIC BUILD TIMELINE

PDF Collection & Extraction		1-2 wks
Data Cleaning & Feature Engg.		1 wk
EDA & Hypothesis Testing		1 wk
ML Modeling & Validation		1-2 wks
Backend API (FastAPI)		1 wk
React Dashboard		1-2 wks
Testing & Deployment		1 wk

PROJECT BUDGET

Python Libraries	₹ 0
React / FastAPI / DB	₹ 0
Cloud Hosting (free tier)	₹ 0
VPS Hosting (optional)	₹ 500-2,000
Domain (optional)	₹ 500-1,000
TOTAL RANGE	₹ 0 - 3,000